9. Let $A = \{1,2,3,4\}$ and $B = \{4,8,9,10\}$. A function $f : A \to B$ given by $f = \{(1,4),(2,8),(3,9),(4,10)\}$ is a

(A) Many-one function (B) Identity function
(C) One-to-one function (D) Into function

10. If $f(x) = 2x^2$ and $g(x) = \frac{1}{3x}$, then $f \circ g$ is

(A) $\frac{3}{2x^2}$ (B) $\frac{2}{3x^2}$ (C) $\frac{2}{9x^2}$ (D) $\frac{1}{6x^2}$

11. If $f : A \to B$ is a bijective function and if $n(B) = 7$, then $n(A)$ is equal to

(A) 7 (B) 49 (C) 1 (D) 14

12. Let f and g be two functions given by

$f = \{(0,1),(2,0),(3,-4),(4,2),(5,7)\}$

$g = \{(0,2),(1,0),(2,4),(-4,2),(7,0)\}$ then the range of $f \circ g$ is

(A) {0,2,3,4,5} (B) {–4,1,0,2,7} (C) {1,2,3,4,5} (D) {0,1,2}

13. Let $f(x) = \sqrt{1+x^2}$ then

(A) $f(xy) = f(x).f(y)$ (B) $f(xy) \geq f(x).f(y)$
(C) $f(xy) \leq f(x).f(y)$ (D) None of these

14. If $g = \{(1,1),(2,3),(3,5),(4,7)\}$ is a function given by $g(x) = \alpha x + \beta$ then the values of α and β are

(A) (–1,2) (B)(2, –1) (C) (–1, –2) (D) (1,2)

15. $f(x) = (x+1)^3 - (x-1)^3$ represents a function which is

(A) linear (B) cubic (C) reciprocal (D) quadratic

Unit Exercise - 1



1. If the ordered pairs $(x^2 - 3x,\ y^2 + 4y)$ and $(-2,5)$ are equal, then find x and y.

2. The cartesian product $A \times A$ has 9 elements among which (–1, 0) and $(0,1)$ are found. Find the set A and the remaining elements of $A \times A$.

3. Given that $f(x) = \begin{cases} \sqrt{x-1} & x \geq 1 \\ 4 & x < 1 \end{cases}$. Find

(i) $f(0)$ (ii) $f(3)$ (iii) $f(a+1)$ in terms of a.(Given that $a \geq 0$)

4. Let $A = \{9,10,11,12,13,14,15,16,17\}$ and let $f : A \to N$ be defined by $f(n) =$ the highest prime factor of $n \in A$. Write f as a set of ordered pairs and find the range of f.

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